A Stock Price Forecasting Model Integrating Complementary Ensemble Empirical Mode Decomposition and Independent Component Analysis

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Abstract

In recent years, due to the non-stationary behavior of data samples, modeling and forecasting the stock price has been challenging for the business community and researchers. In order to address these mentioned issues, enhanced machine learning algorithms can be employed to establish stock forecasting algorithms. Accordingly, introducing the idea of "decomposition and ensemble" and the theory of "granular computing", a hybrid model in this paper is established by incorporating the complementary ensemble empirical mode decomposition (CEEMD), sample entropy (SE), independent component analysis (ICA), particle swarm optimization (PSO), and long short-term memory (LSTM). First, aiming at reducing the complexity of the original data of stock price, the CEEMD approach decomposes the data into different intrinsic mode functions (IMFs). To alleviate the cumulative error of IMFs, SE is performed to restructure the IMFs. Second, the ICA technique separates IMFs, describing the internal foundation structure. Finally, the

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Computational Neuroscience | Computational Economics | Computer Science | Financial Econometrics | Information Model | Quantitative Finance | Financial Time Series Prediction Using Machine Learning Techniques to predict the stock market fluctuations. The Markov Switching method was then conducted to specify the states of external effects. Finally, the experimental results indicated that the GARCH model with external factors could improve the forecasting accuracy. Pai and Lin [7] utilized the ARIMA model to capture the linear patterns. The data results revealed that the ARIMA model could utilize the unique strength of stock price trend forecasting.

Due to the nonlinear behavior and fast variations of stock prices, the statistical models' ability to forecast the stock price is limited [9,10,11]. With the rapid growth of artificial intelligence, several efficient artificial intelligence approaches with strong robustness and fault tolerance have been presented for predicting the stock price in the literature, such as artificial neural networks (ANNs) (Elman neural network (ENN) [3], generalized regression neural network(GRNN) [12], radial basis function network (RBFN) [13] and wavelet neural network (WNN) [3], fuzzy logic methods [14], support vector machine (SVM) [15, 16], least-squares support vector machine (LSSVM) [17], and extreme learning machine (ELM) [18].

combining the financial factors and ARIMAX-GARCH family models for forecasting the stock price index. Compared with the benchmark models, the results reflected an enhancement in the hybrid model's stock index prediction. Kristjanpoller et al. [6] combined ANFIS and GARCH techniques to specify individual effects on any stock index and used an ANN framework to enhance the stock price's prediction performance. The results indicated that the mentioned approach could better estimate the stock index fluctuations.

The artificial intelligence method is prevalent in the stock field, but it is unstable, and parameter initialization can affect its results [24,25,26]. Furthermore, to solve the instability problem of the ANNs, various optimization algorithms, such as particle swarm optimization (PSO) [3, 27], firefly algorithm [28], whale optimization algorithm (WOA) [29, 30], and an enhanced version of the cuckoo search algorithm (CSA) [31], have been utilized to optimize the ANNs' initial weight and threshold. Chandar [32] employed the NN optimized by genetic algorithms and PSO for forecasting the intraday stock price. The results

Based on the mentioned analysis above, there are two disadvantages in the existing stock price prediction methods. On the one hand, the internal signal of the original data can not be revealed. On the other hand, the IMFs obtained by decomposition method have cumulative error. Therefore, aiming at solving these problems, this paper proposes a new prediction framework of stock price based on the theory of "granular computing" and the idea of "decomposition and integration". First, the collected unstable stock price data are processed by decomposition method to obtain the subsequences with different characteristics. Then the complexity measurement method is used to calculate the entropy of the subsequences. Meanwhile, for the sake of alleviating the cumulative error, the sequences with similar entropy are merged. Next, the essential characteristics of stock price data is revealed. Finally, the neural network improved by the optimization algorithm is utilized to forecast the merged subsequences, respectively. Finally, the forecasting results are integrated. Specifically, the stock price forecasting model based on ICA and CEEMD is established to reveal the primitive characteristics of stock price data and construct a more efficient hybrid

2.1 CEEMD

EMD decomposition has been extensively utilized to transform the complex initial series signal with nonlinear and nonstationary characteristics into a class of fluctuating modes described with IMFs, assuming that each signal can be a mix of IMFs series. There exist two conditions for any IMF: the extrema and zero-crossings should have similar values throughout the dataset, and the summation of the peak and valley values should be zero. The second condition indicates that the positions of upper and lower envelopes are symmetrically distributed along with a zero axis. Nevertheless, modal aliasing may be caused by the EMD approach, resolved through noise-based analysis. In terms of the average frequency distribution feature of Gaussian white noise added to the series, the signal has continuity on various scales, avoiding modal aliasing to a certain extent [38]. In order to prevent the mode mixing in EMD, the EEMD is further enhanced using a random Gaussian white noise incorporated with the original series data [39]. Yeh et al. [40] extended EMD to EEMD and CEEMD. Time series can be reestablished to eliminate the residual auxiliary noise and attain

The results of any IMF are generated after taking an average of the entire ensembles, as described in the following:

$$\begin{aligned} I M F_{j}=\frac{1}{2 N} \sum _{i=1}^{2 N} I M F_{i j} \end{aligned}$$

(2)

Now, the final signal $Y(t)$ can be generated after applying CEEMD as:

$$\begin{aligned} Y(t)=\sum _{j=1}^{k} {IMF}_{j}(t)+r e s \end{aligned}$$

(3)

where *res* is the residual term.

(5)

where X stands for the estimated value of S. Since ICA includes the uncertainties of variances and order, y_i is not necessarily an estimation of s_i. The following three assumptions can be considered: the first one is that the mixing matrix A is full rank, the second is that source signals are mutually independent, and the third is that there is at most an individual source signal with a Gaussian distribution.

3 CEEMD and ICA-Based Stock Price Forecasting Model

The current section adopts a decomposition-hybridization algorithm based on the CEEMD, SE, ICA, PSO, and LSTM methods to forecast the stock price, abbreviated to CS-ICA-PSO-LSTM. The following subsection presents various performance measures to attain the

Boundary impact and mode mixing are the main shortcomings of the decomposed IMFs. In order to resolve the mentioned difficulties, the stock price is decomposed into various IMFs using the CEEMD [40]. Based on CEEMD, the raw time series of stock price can be decomposed into different components. Because CEEMD decomposition method transforms complex time series into relatively stable and regular IMFs, the hidden information is easy to mine. The detailed process and definition are presented as following:

Definition 3.1

Let (U, E, I, F) be a hybrid forecasting system. Assume ε denote the decomposition error. The relation of U and I is defined as following:

$$\begin{aligned} x_i-IMF_{ij}\le \varepsilon \quad or \quad X-IMF_{j=1}^{m}\le \varepsilon (i=1, 2, \ldots , n) \end{aligned}$$

\end{array}\right] \end{aligned}$$

(7)

(2) The distance between vectors \(\mathbf {y}(i)\) and \(\mathbf {y}(j)\), denoted by \(d[\mathbf {y}(i),\mathbf {y}(j)]\), is described as.

$$\begin{aligned}&d[\mathbf {y}(i), \mathbf {y}(j)]=\max (\mid \mathbf {y}(i+k)-\mathbf {y}(j+k) \mid)\nonumber \\&\quad (1 \le k \le m-1 ; 1 \le i \ne j \le n-m+1) \end{aligned}$$

(8)

where *k* stands for the step length.

(3) Considering a threshold r, if \(B_i\) is a number for a specific \(\mathbf {y}(i)\)

(6) Obtain the SE value as:

$$\begin{aligned} E(m, r)=\lim _{n \rightarrow \infty }\left\{ -\ln \left[\frac{B^{m}(r)}{B^{m+1}(r)}\right] \right\} \end{aligned}$$

(12)

The SE values determine the series' nonstationary degree; that is, the series complexity increases with the increase of the SE value.

3.2 Independent Component Extraction and Prediction

The revelation of source signals is the fundamental stage for establishing a stock price forecasting model. As a commonly used approach to determine the measured data's underlying parameters, ICA has been introduced into the revelation of source signals [41]. It

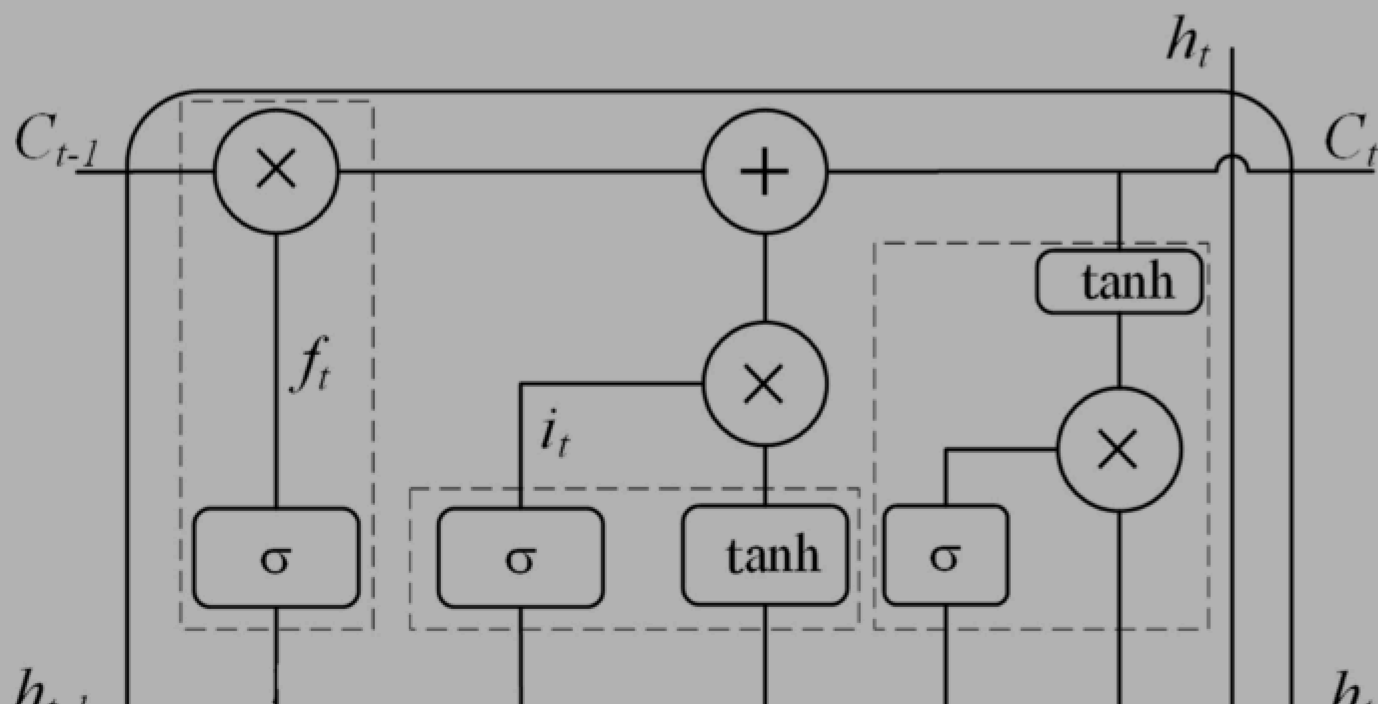
Finally, the LSTM is modeled on the residual ICs to establish a stock price prediction model. Notably, the LSTM's prediction accuracy depends on determining the hyperparameters. In order to attain accurate forecasting results, the PSO [45] algorithm is adopted to find the optimum values of the LSTM hyperparameters. The PSO, first presented by Eberhart et al. [46] in 1995, is one of swarm intelligent optimization algorithms. This algorithm originated from the study of bird predation. The simplest and most effective way of finding edibles is to search for the nearest surrounding area where the birds have found food. Inspired by birds (particles) flocks looking for food, PSO employs the shared data between the particles to determine the global optimal solution. After initializing the particles, an iterative procedure is adopted to find the global optimal solution. In any iteration, the *i*th particle employs the particle tracking format at the optimal position of individual p_{best} and swarms g_{best} to update its current position pi and velocity v_i, respectively. This means that the v_i and p_i are updated. See [47, 48] to better understand the PSO algorithm and its applications.

LSTM, as a refined version of the RNN structure, was employed to resolve the gradient vanishing problems of RNN [49]. The LSTM employs a gating approach to control adding or removing the cells' state information. The cell state is equivalent to an information transmission path and is implemented immediately on the whole chain. The gate framework selectively allows the information to pass. Figure 1 describes the structure of an individual LSTM module.

In the LSTM structure, C_{t-1} and h_{t-1} represent the cell state and output of the upper layer. y_t refers to the new input. $\tanh$ is the generated new cell value. Three σ symbols are the input, output, and forgetting gates, respectively. These three gates cooperate to control the whole cell state. The renewal process of each cell is as follows:

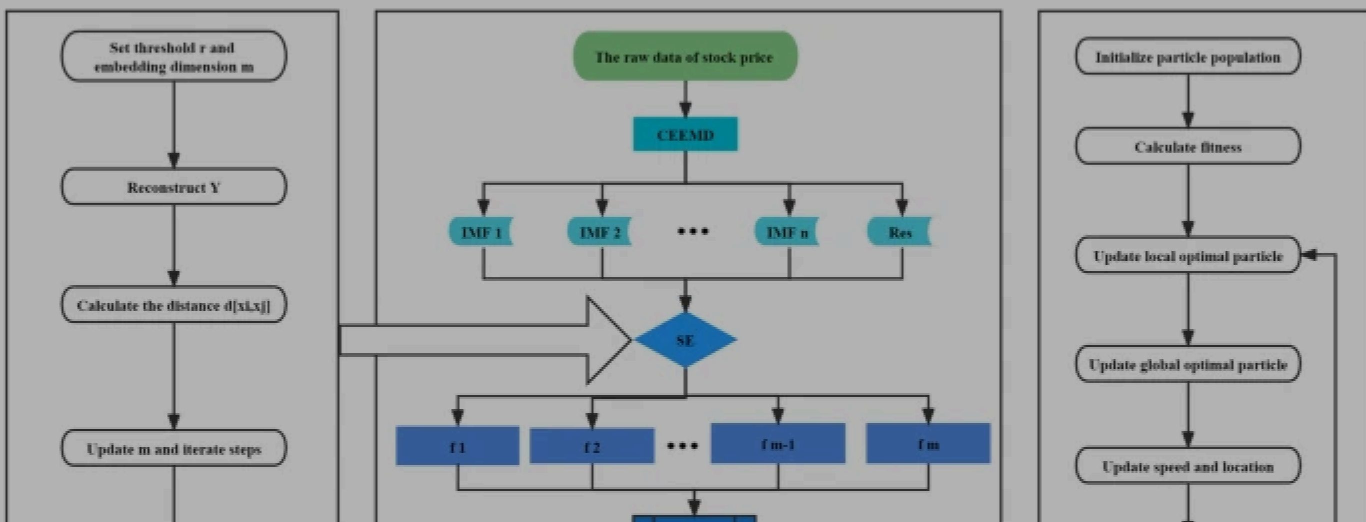
(1) Let W_f and b_f represent the weights and bias. The following formula determines all information to be abandoned in the forgetting gate:

Fig. 1



LSTM optimized by PSO is utilized to forecast the all independent components. These parameters of all algorithms are listed in Table 1.

Fig. 2



$$\begin{aligned}&M A P E=\frac{1}{n} \sum _{i=1}^{n} \mid \frac{Y_{i}-\hat{Y}_{i}}{Y_{i}} \mid \times 100 \% \end{aligned}$$

(19)

$$\begin{aligned}&M A E=\frac{1}{n} \sum _{i=1}^{n} \mid Y_{i}-\hat{Y}_{i} \mid \end{aligned}$$

(20)

$$\begin{aligned}&R M S E=\sqrt{\frac{1}{n} \sum _{i=1}^{n}\left(Y_{i}-\hat{Y}_i\right) ^{2}} \end{aligned}$$

(21)

where $d_{i}=\left\{ \begin{array}{ll} 1, &{} \left(Y_{i}-Y_{i-1}\right) \left(\hat{Y}_{i}-Y_{i-1}\right) \ge 0 \\ 0, &{} { otherwise } \end{array}\right.$; n stands for the whole number of data points; Y_i and $\hat{Y}_i$ describe the actual and predicted values of the ith term, respectively.

MAPE, MAE, and RMSE are employed to determine the difference between the predicted and actual values. The lower values of the mentioned measures indicate a superior prediction accuracy. Nevertheless, better prediction accuracy can be obtained for higher values of the *DA* and R^2. In addition, Shannon entropy is utilized to measure the complexity of neural networks, which is expressed as the following:

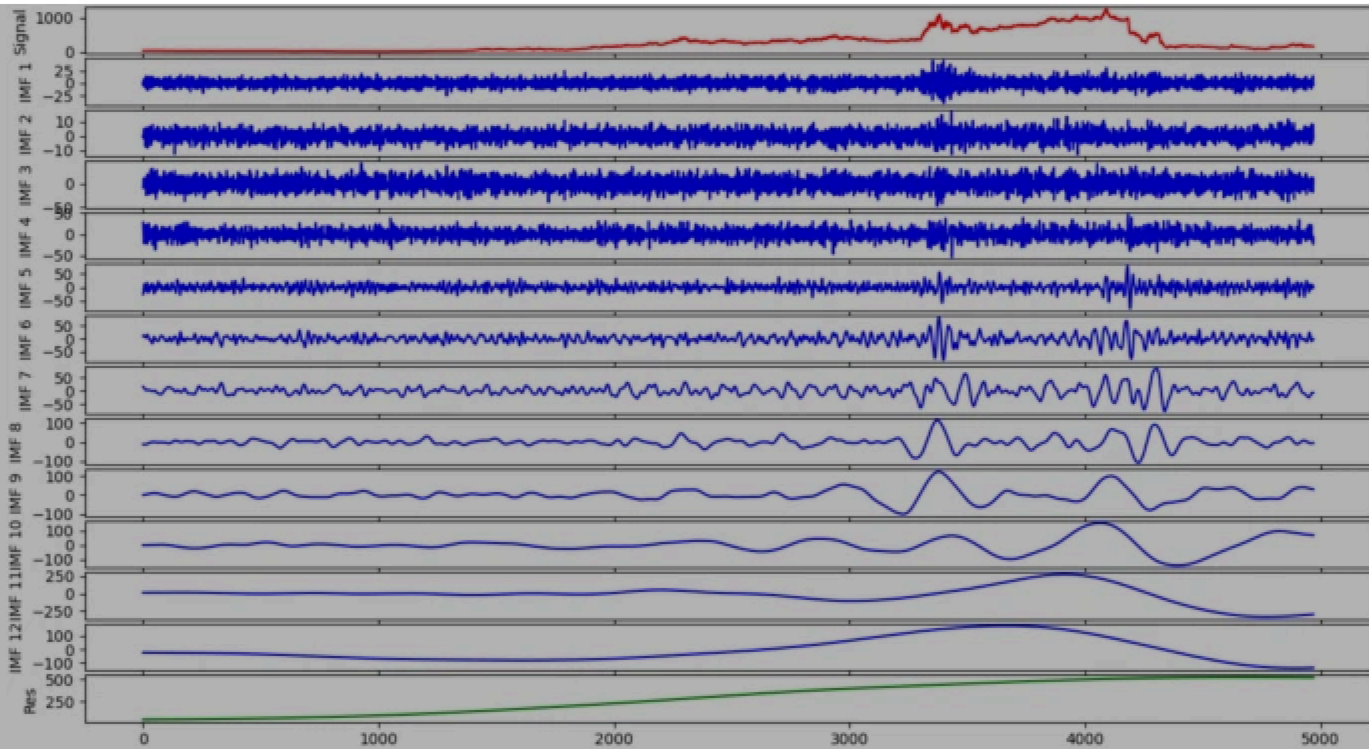
$$\begin{aligned} S_e=-k \sum _{i=1}^{N} p_{i}=\frac{ \text{ D } (i)}{\sum _{i=1}^{N}

chooses four stocks in the SSE 50 index as candidate assets. The ticker symbols of four stocks are “SH600518”, “SH600519”, “SH600999”, and “SH601988”. The statistical description about the 4 stocks is presented as following:

Table 2 The statistical properties of stocks

Data from March 19, 2001 to March 16, 2021 are collected and divided into a training set and a testing set. The training set ranges from March 19, 2001 to April 17, 2017 and the testing set is from April 18, 2017 to March 16, 2021. The ratio of training set and testing set is 8:2.

4.2 The CEEMD Decomposition and Reconstruction Results

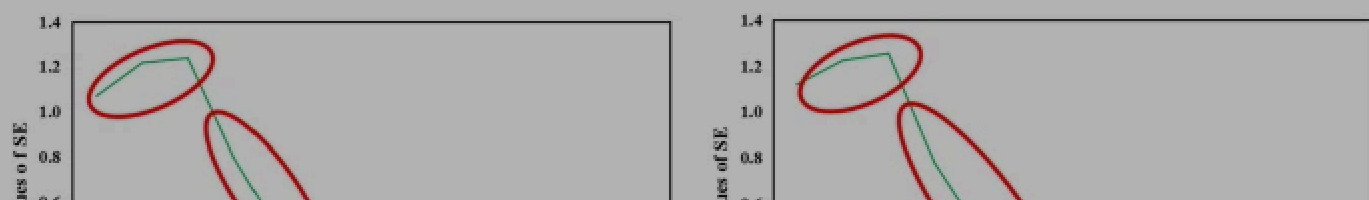


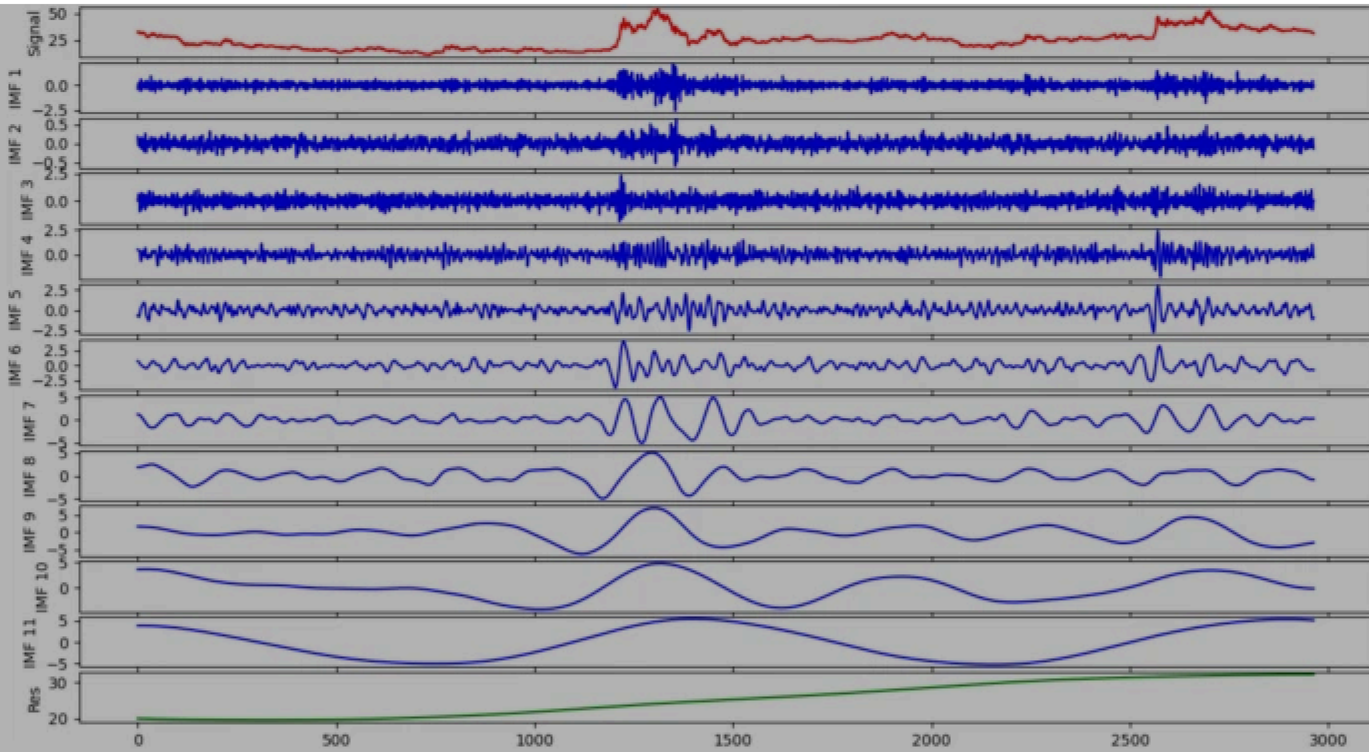
The decomposition results of CEEMD for SH600518

The decomposition results of CEEMD for SH600519

Table 3 The values of sample entropy of IMFs

Fig. 5





The decomposition results of CEEMD for SH600999

The decomposition results of CEEMD for SH601988

4.3 The ICA Results

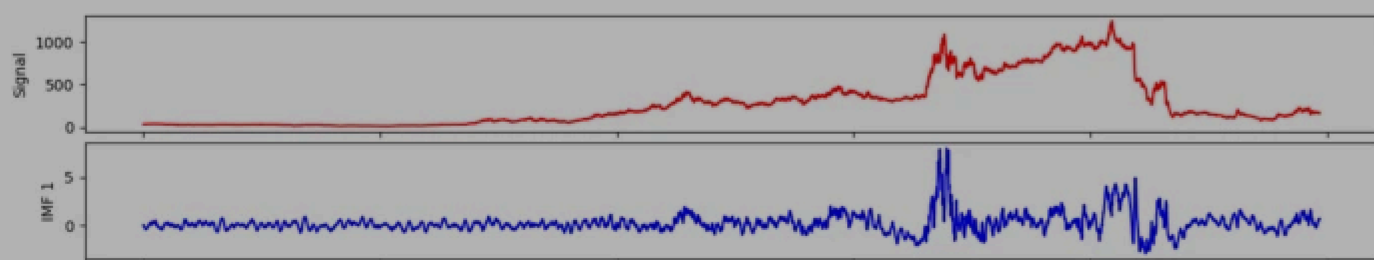
The separation unit is another step of the established model. We start with separating ICs. The ICA model is applied to the IMFs to attain ICs. Moreover, the ICs are found by revealing the source signal. Finally, the ICs with source signals are considered as the input of the LSTM for the construction of the forecasting model.

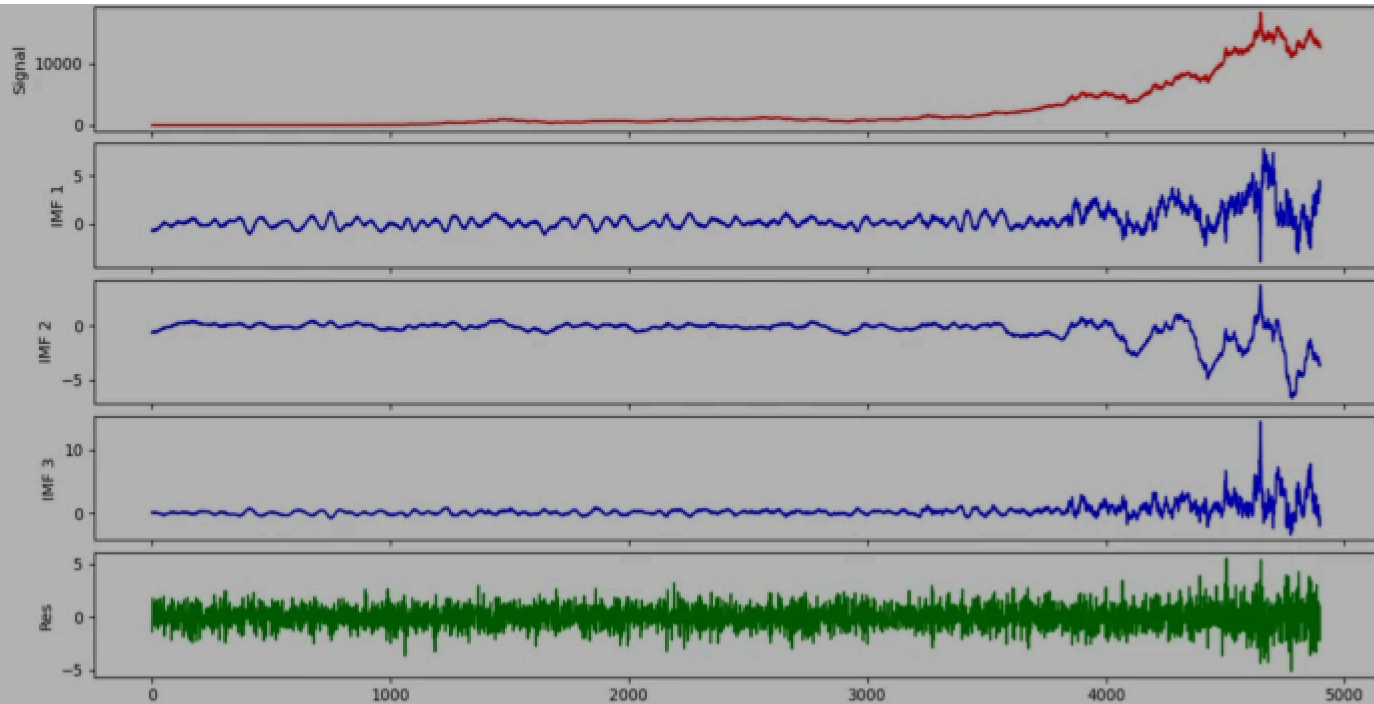
The four ICs are separated through the ICA from the recombination modes. Figs. 8, 9, 10, 11 describe the final results obtained with the enhanced CS-ICA method. Based on Eq. (13), four separation matrixes are derived.

$$\begin{aligned}&M_{S H 601988}=\left(\begin{array}{cccc} 0.9990 & -0.0078 & -0.0004 & 0.0438 \\ -0.0044 & 0.6854 & 0.6910 & 0.2294 \\ 0.0149 & 0.7281 & -0.6518 & -0.2142 \\ 0.0417 & 0.0010 & 0.3141 & -0.9485 \end{array}\right) \end{aligned}$$

(28)

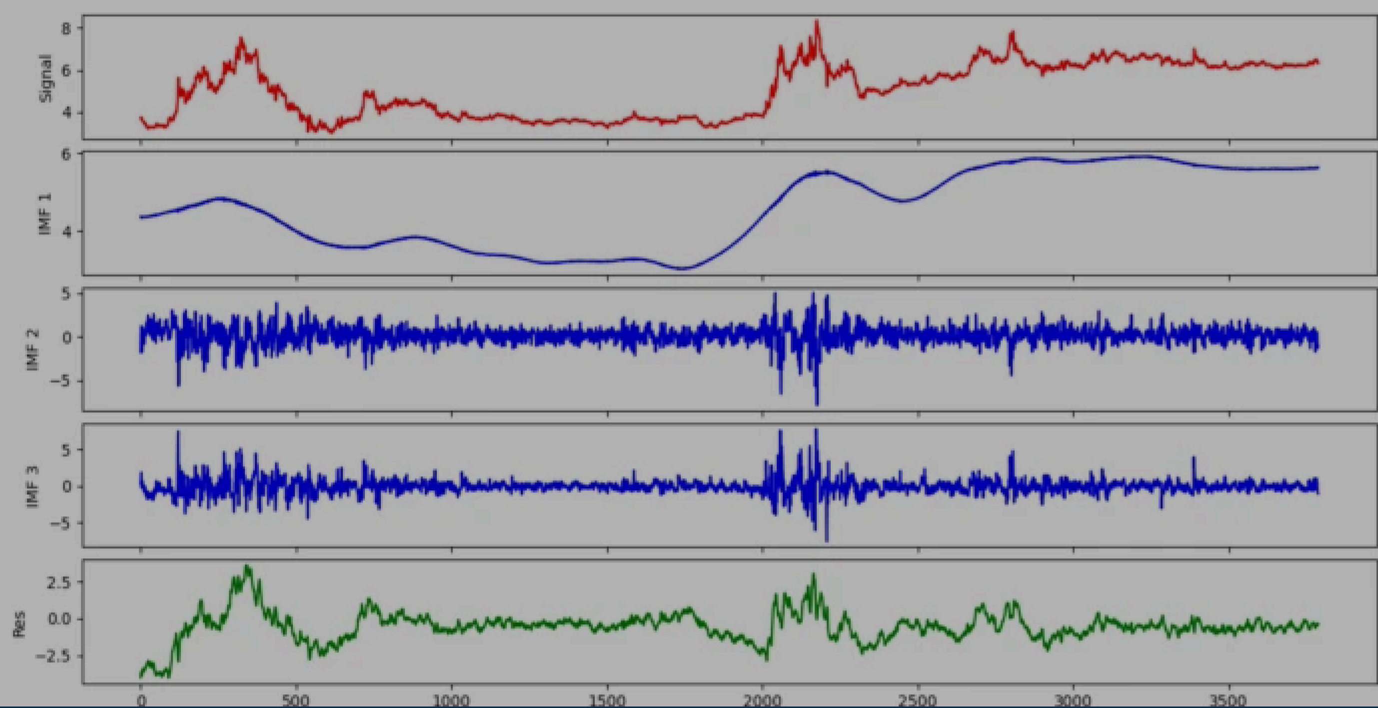
Fig. 8





The results of ICA for SH600519

Fig. 11



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$$\begin{aligned}&\hat{x}(t)_{S H 600999}=\sum _{i=1}^{4} a_{i} s_{i}(t)=0.3120 s_{1}+1.3016 s_{2}\nonumber \\&\quad\quad\quad\quad\quad +1.2400 s_{3}+0.8192 s_{4} \end{aligned}$$

(31)

$$\begin{aligned}&\hat{x}(t)_{S H 601988}=\sum _{i=1}^{4} a_{i} s_{i}(t)=1.0346 s_{1}+1.6014 s_{2}\nonumber \\&\quad\quad\quad\quad\quad -0.1222 s_{3}-0.5917 s_{4} \end{aligned}$$

(32)

4.4 Comparing the Prediction Accuracy

In order to verify the accuracy of the established model, the current work employs some

SH600999, the MAE, RMSE, and MAPE values of PSO-LSTM for SH600999 are 45.07%, 35.91, and 43.15%, respectively. Hence, it has been verified that the PSO model is more suitable for stock forecasting. Compared with the ARIMA model, it is evident that the LSTM model provides better efficiency in 1-step prediction. For instance, Table 5 shows that the MAE, RMSE, MAPE, and DA values of the PSO-LSTM are decreased by 18.33%, 70.91%, 74.14%, and 0.09% than ARIMA in 1-step prediction, respectively. According to these mentioned analysis, the PSO approach providing optimal parameters for LSTM achieves a better forecasting performance than the LSTM model.

(3) Compared with PSO-LSTM, CS-PSO-LSTM provides higher prediction accuracy. For example, the MAE, RMSE, and MAPE of CS-PSO-LSTM at 1-step forecasting are increased by 41.67%, 31.25%, and 36.11% for SH601988, respectively. The results indicate that the CEEMD algorithm optimized by SE is an improved decomposition algorithm. Compared with the ARIMA, BP, and LSTM, the increase in MAPE of the CS-PSO-LSTM model for SH601988 in the 3-step forecasting is 80.60%, 78.37%, and

Table 4 Forecasting performance of different models for SH600518

Table 5 Forecasting performance of different models for SH600519

Table 6 Forecasting performance of different models for SH600999

The proposed ES-ICA-PSO-LSTM model comprises various models, including the CEEMD, SE, ICA, PSO, and LSTM, to forecast the stock price. The outstanding performance is attributed to the following reasons: (1) Based on the theory of "granular computing" and "decomposition and ensemble", the raw data of stock price are decomposed into different components. On the one hand, the hidden information is revealed. On the other hand, different features (trend, period, random, etc.) are classified. (2) SE is conducted to restructure the IMFs and alleviate the cumulative error. (3) The ICA technique can describe the internal foundation structure of IMFs, which is key to reveal the essence of the original signal. In theory, a useful attempt is made by integrating the idea of "granular computing" with "decomposition and ensemble" to construct the forecasting model of non-stationary data. In practice, the research results will provide scientific reference for the business community and researchers.

The stock prices affected by emergencies is difficult to accurately forecast. Future aspects are

CEEMD: Complementary ensemble empirical mode decomposition

CS: CEEMD and SE

CSA: Cuckoo search algorithm

DA: Directional accuracy

ENN: Elman neural network

ELM: Extreme learning machine

EMD: Empirical mode decomposition

PSO: Particle swarm optimization

RBFN: Radial basis function network

RMSE: Root mean square error

SE: Sample entropy

SSA: Singular spec tral analysis

SVM: Support vector machine

VMD: Variational mode decomposition

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Contributions

YC completed the main work of this paper under the guidance of PZ, containing methodology and writing original draft. ZZ performed the validation and formal analysis. JB and YG implemented the editing work. All authors read and approved the final manuscript.

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Ethics declarations

Conflict of Interest

All authors declare that they have no conflicts of interest.

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Keywords

Stock price forecasting | Complementary ensemble empirical mode decomposition | Independent component analysis | Long short-term memory